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A robust filter in stock networks analysis

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ABSTRACT

We show that the use of a minimal spanning tree (MST) to filter important information in a complex system is not robust except when the system contains a unique MST. In this paper we propose to use the forest of all MSTs as a robust filter. According to this filter, centrality measures are also robust. For that purpose an algorithm, which can also be used to detect the uniqueness of an MST, will be provided. A simple hypothetical example will clarify the construction of the proposed filter and a real problem in filtering the information contained in NYSE 100 stocks will illustrate its advantages compared to the MST-based filter.

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1. Introduction

Since it was introduced a decade ago by Mantegna [1], minimal spanning trees (MSTs) have become an indispensable tool in econophysics to filter important information contained in the complex structure of a correlation matrix among stocks in a given portfolio [2,3]. See also, for example, [4] for further discussion on filtering information in complex systems and [5,6] for the application of MSTs to investigate the degree distribution of individual stocks listed on the S&P500 and KOSPI200. More specifically, MSTs could help to explain the physical correlation between the topological structure and economic classification of the stocks [7]. If the filtered topological structure of the stocks could be directly obtained from the MST, the economic classification in the form of an indexed hierarchical tree is constructed based on the subdominant ultrametric (SDU) related to that MST [1,3]. However, as we will show, the use of an MST to filter economic information in a network between stocks might produce misleading information if there are more than one MSTs contained in the network. This is what we intend to handle.

In practice, the MST that is used to analyze a network between stocks is usually obtained from the two most suggested algorithms in the literature, i.e., Kruskal's algorithm and Prim's algorithm (see e.g., [1,3,8]). The important role of those algorithms in the construction of filtered network topology and economic taxonomy of stocks motivated Huang et al. [9] to compare them to each other in terms of computational complexity. They reported the conditions where one algorithm is superior to the other. Unfortunately, both algorithms provide only one MST among all MSTs that might possibly exist in the network and cannot detect the uniqueness of the MST. These restrictions indicate that there is also an algorithmic problem to handle.

The objective of this paper is to propose to filter the information in stock networks based on the forest of all MSTs (or briefly the 'forest') instead of based on one MST. Although the MST might not be unique in the network, the forest is unique. In other words, there might be many different possible network topologies that can be used if one works based on an MST but there is only one network topology if one works with the forest. That uniqueness guarantees the robustness of the filtered information provided by the forest. In order to determine that forest, an algorithm will be developed, two

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illustrative examples will be presented, and the stock's degree centrality will be discussed to illustrate its advantage. The rest of the paper is organized as follows. In the next section, a distance matrix D will be considered as a fuzzy relation to construct the min–max transitive closure of stock networks. Later, in Section 3, we show that it is the SDU of D . An algorithm to find the forest of all MSTs will be presented in Section 4 followed in Section 5 by (i) a hypothetical example to show how that algorithm works in practice and (ii) a real example using NYSE 100 daily stock prices to illustrate the advantages of the proposed robust filter. A discussion on degree centrality in Section 6 is to show one of those advantages. Concluding remarks will close the presentation of this paper.

2. Min–max transitive closure of stock networks

Let E be a set of n stocks in a portfolio under study. In stock networks analysis, the interrelations among stocks are represented in terms of correlation. Let $p_i(t)$ and $r_i(t)$ be the price of stock i and the logarithm of i -th stock return at time t , respectively. Thus,

$$r_i(t) = \ln p_i(t) - \ln p_i(t-1)$$

for all $i = 1, 2, \dots, n$. The interrelations among stocks are usually summarized in the form of a correlation matrix C of size $(n \times n)$, where the element of the i -th row and j -th column is,

$$c(i, j) = \frac{\langle r_i r_j \rangle - \langle r_i \rangle \langle r_j \rangle}{\sqrt{\langle r_i^2 \rangle - \langle r_i \rangle^2} \sqrt{\langle r_j^2 \rangle - \langle r_j \rangle^2}}$$

with $\langle r_i \rangle$ being the average of $r_i(t)$ for all t [3,10]. In other words, the matrix C is a numerical representation of the complex system of stocks' interrelationships.

To analyze these interrelationships, it is customary to transform C into a distance matrix D , where its i -th row and j -th column is

$$d(i, j) = \sqrt{2 \{1 - c(i, j)\}}$$

for all $i, j = 1, 2, \dots, n$. Then, the network between stocks is analyzed through the network topology created by an MST of D [3]. If D contains a unique MST, the filtered information provided by the MST is robust in the sense that there is no other source of filtered information that can be used. However, in practice, it is very often that D contains more than one MST. See, for example, the case of NYSE 100 stocks discussed in Section 5. In this case, the use of an MST is not robust. This motivates us to use the forest of all MSTs as a robust filter. For that purpose, in what follows we develop an algorithm to construct that forest by using a fuzzy relation approach. This approach allows us to see further the properties of D .

As a real non-negative function defined on $E \times E$, d satisfies

- (i) $0 \leq d(i, j) < \infty$ and $d(i, j) = 0$ if and only if $i = j$,
- (ii) $d(i, j) = d(j, i)$, and
- (iii) $d(i, j) \leq d(i, k) + d(k, j)$,

for all i, j and k in E . From the fuzzy relation viewpoint, these properties imply that D is a symmetric and anti-reflexive fuzzy relation with d as the membership function [11]. Therefore, if we consider the k times min–max transitive operation $*$ on D and itself,

$$D^{*k} = D * D^{*(k-1)} \quad \text{for all } k = 2, 3, \dots$$

where

- (i) $D^{*1} = D$,
- (ii) the membership function d^{*k} in D^{*k} is

$$d^{*k}(i, j) = \bigwedge_{m=1}^n \{d(i, m) \vee d^{*(k-1)}(m, j)\}, \quad (1)$$

- (iii) $a \wedge b = \min\{a, b\}$ and $a \vee b = \max\{a, b\}$ for all real numbers a and b ,
- then we have the following property.

Theorem 1. The sequence $D, D^{*2}, D^{*3}, \dots, D^{*k}$ is monotone decreasing, i.e.,

$$D^{*k} \subseteq \dots \subseteq D^{*3} \subseteq D^{*2} \subseteq D.$$

Proof. By definition, from (1),

$$d^{*2}(i, j) = \bigwedge_{k=1}^n \{d(i, k) \vee d(k, j)\}.$$

For $k = j$, we get

$$d^{*2}(i, j) \leq d(i, j) \vee d(j, j) = d(i, j).$$

This implies $D^{*2} \subseteq D$. Therefore, by proceeding in a similar manner, in general, we have

$$D^{*k} \subseteq \dots \subseteq D^{*3} \subseteq D^{*2} \subseteq D, \quad (2)$$

which proves the theorem. $\square$

Corollary 1. The min–max transitive closure D^+ of D , defined in Ref. [11] as

$$D^+ = D \cap D^{*2} \cap D^{*3} \cap \dots \cap D^{*k},$$

is simply $D^+ = D^{*k}$.

From this corollary we learn that min–max transitive closure of D is the k times min–max operation ** on D for a positive integer k . If $d^+(i, j)$ is the i -th row and j -th column of D^+ , from (2), $d^+(i, j) \leq d(i, j)$ for all pairs (i, j) . In the next section we show that D^+ is the SDU of D , i.e.,

$$d^+(i, j) \leq \sup_{\delta} \{ \delta(i, j) : \delta \text{ is an ultrametric distance and } \delta(i, j) \leq d(i, j) \}$$

for all i and j in E .

3. Construction of the SDU

First, we present the following theorem, which is a dual form of Theorem II in Ref. [11] page 95 for the case of the fuzzy relation D . By dual form we mean that, if that theorem in Ref. [11] is about max–min transitive closure, the theorem below is about min–max transitive closure.

Theorem 2. Let d^+ be the membership function of D^+ . Then, $d^+(i, j) = \bigwedge_{\gamma \text{ in } \Gamma} L(\gamma)$ for all pairs (i, j) in $E \times E$, where

- (i) $\Gamma = \{ \gamma \mid \gamma = (i = i_1, i_2, \dots, i_p = j) \text{ is a chain from } i \text{ to } j \}$,
- (ii) $L(\gamma) = \bigvee_{k=1}^{p-1} d(i_k, i_{k+1})$.

In this theorem $\bigvee_{k=1}^{p-1} d(i_k, i_{k+1})$ is the length of the chain γ and $d^+(i, j)$ is the length of the shortest chain from i to j . Those who are familiar with cluster analysis will recognize that the technique of hierarchical clustering based on the construction of D^+ is the so-called single linkage. Based on this theorem and the corollary of Theorem 1, in what follows we show that D^+ is equal to the SDU of D . This property will lead us to the construction of the forest of all MSTs in D in the next section.

Theorem 3. D^+ is an ultrametric distance matrix.

Proof. Consider again $\gamma = (i = i_1, i_2, \dots, i_p = j)$ and $L(\gamma)$ defined in Theorem 2. Let γ_1 be a chain from i to k and γ_2 be a chain from k to j such that $d^+(i, k) = L(\gamma_1)$ and $d^+(k, j) = L(\gamma_2)$. Suppose γ_3 is the chain from i to j such that $L(\gamma_3) = L(\gamma_1) \vee L(\gamma_2)$. Such γ_1, γ_2 and γ_3 always exist because D is a connected graph. Then,

$$L(\gamma_3) = d^+(i, k) \vee d^+(k, j).$$

But, from Theorem 2, $d^+(i, j) = \bigwedge_{\gamma \text{ in } \Gamma} L(\gamma)$. Therefore, $d^+(i, j) \leq L(\gamma_3)$ or, equivalently, $d^+(i, j) \leq d^+(i, k) \vee d^+(k, j)$. This means that d^+ is an ultrametric distance. Thus, the proof is completed. $\square$

Theorem 4. D^+ is the SDU of D .

Proof. Suppose U is the SDU of D . Since the membership function u of U is an ultrametric distance, it satisfies $u(i, j) \leq u(i, k) \vee u(k, j)$ for all i, j and k in E . In particular,

$$u(i, j) \leq u(i_1, i_k) \vee u(i_k, i_p)$$

for all $k = 2, 3, \dots, (p - 1)$. Hence,

$$u(i, j) \leq \bigvee_{k=1}^{p-1} u(i_k, i_{k+1}).$$

Since U is the SDU of D , which means $u(i, j) \leq d(i, j)$ for all i and j in E , we have

$$u(i, j) \leq \bigvee_{k=1}^{p-1} d(i_k, i_{k+1}).$$

Therefore,

$$u(i, j) \leq \bigwedge_{\gamma \text{ in } \Gamma} \bigvee_{k=1}^{p-1} d(i_k, i_{k+1}) = L(\gamma) = d^+(i, j). \quad (3)$$

Furthermore, since D^+ is an ultrametric and U is the SDU of D , the inequality (3) implies $u(i, j) = d^+(i, j)$. Thus, $U = D^+$.

We conclude that, to construct SDU of D is to find D^+ and vice versa. In the next section, we exploit this property of D^+ to construct an algorithm for finding the forest of all MSTs. $\square$

4. An algorithm

The SDU of D is unique but the MST of D might not be. Through the notion of SDU, in this section we discuss the construction of the forest of all MSTs. Let M be an MST of D and $(i = i_1, i_2, \dots, i_p = j)$ be the chain from i to j in M . If we define the distance $\hat{d}$ between i and j in M by

$$\hat{d}(i, j) = \bigwedge_{\gamma \text{ in } \Gamma} \bigvee_{k=1}^{p-1} d(i_k, i_{k+1})$$

then, according to (3), $\hat{d}$ is the SDU of d . Therefore, there exists an integer m ; $1 \leq m < p$, and a γ in Γ such that

$$d^+(i, j) = \hat{d}(i, j) = d(i_m, i_{m+1}).$$

As a consequence, if Δ is a fuzzy relation where its membership function is given by

$$\delta(i, j) = \begin{cases} 1; & d(i, j) - d^+(i, j) = 0 \quad \text{and} \quad i \neq j \\ 0; & d(i, j) - d^+(i, j) \neq 0 \quad \text{or} \quad i = j, \end{cases} \quad (4)$$

then Δ is the adjacency matrix that corresponds to the forest of all MSTs. In other words, that forest is defined by all pairs (i, j) where $i > j$ and $\delta(i, j) = 1$. Furthermore, by observing the number of those pairs, we obtain a necessary and sufficient condition for the uniqueness of MST in D . This is formulated in the following theorem (see also [12]).

Theorem 5. Let N be the number of pairs (i, j) where $i > j$ and $\delta(i, j) = 1$. Then, MST in D is unique if and only if $N = (n - 1)$.

This is an important property to check whether the MST in the stock networks under study is unique or not.

Based on the results presented above, we propose the following algorithm to find the forest of all MSTs. In this algorithm, D and Δ are considered as matrices of size $(n \times n)$.

Step 1. Let $k = 2$,

Step 2. Compute D^{*k} , where $D * D^{*(k-1)}$ is a matrix multiplication in the usual sense but multiplication and summation of two real numbers a and b are defined as $\max\{a, b\}$ and $\min\{a, b\}$, respectively,

Step 3. If $D^{*k} = D^{*(k-1)}$, then the SDU of D is D^{*k} and go to Step 4. Otherwise, let $k := k + 1$ and then go back to Step 2,

Step 4. Compute Δ as defined in (4). Then Δ is the adjacency matrix representing the forest of all MSTs in D .

The convergence speed of this algorithm can be increased by computing directly $D^{*2}, D^{*4}, D^{*8}, \dots$, instead of $D^{*2}, D^{*3}, D^{*4}, \dots$. This computation process is stopped at the k -th iteration if $D^{*2^k} = D^{*2^{(k-1)}}$. Then, D^{*2^k} is the SDU of D . In this case, the number of iterations needed is k satisfying $2^k \leq n$, i.e., $k \leq \frac{\ln(n)}{\ln(2)}$.

5. Illustrative examples

Two illustrative examples will be presented here. The first example is to show how the above algorithm works by using a simple hypothetical network. The second one is to present the advantages of the proposed robust filter, compared to the MST-based filter, by using NYSE 100 daily stock prices.

5.1. A hypothetical example

We consider the following distance matrix D representing the hypothetical network in Fig. 1. The element $d(i, j)$ represents the distance between the nodes i and j ; $i, j = A, B, C, D$, and E .

$$D = \begin{pmatrix} 0 & 0.1 & 0.3 & 0.2 & 0.1 \\ 0.1 & 0 & 0.3 & 0.4 & 1.7 \\ 0.3 & 0.3 & 0 & 0.6 & 0.5 \\ 0.2 & 0.4 & 0.6 & 0 & 0.2 \\ 0.1 & 1.7 & 0.5 & 0.2 & 0 \end{pmatrix}.$$

The SDU of D and the adjacency matrix Δ are computed as follows.

Step 1. Computing D^{*2} ,

$$\begin{aligned} D^{*2} &= \begin{pmatrix} 0 & 0.1 & 0.3 & 0.2 & 0.1 \\ 0.1 & 0 & 0.3 & 0.4 & 1.7 \\ 0.3 & 0.3 & 0 & 0.6 & 0.5 \\ 0.2 & 0.4 & 0.6 & 0 & 0.2 \\ 0.1 & 1.7 & 0.5 & 0.2 & 0 \end{pmatrix} * \begin{pmatrix} 0 & 0.1 & 0.3 & 0.2 & 0.1 \\ 0.1 & 0 & 0.3 & 0.4 & 1.7 \\ 0.3 & 0.3 & 0 & 0.6 & 0.5 \\ 0.2 & 0.4 & 0.6 & 0 & 0.2 \\ 0.1 & 1.7 & 0.5 & 0.2 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0.1 & 0.3 & 0.2 & 0.1 \\ 0.1 & 0 & 0.3 & 0.2 & 0.1 \\ 0.3 & 0.3 & 0 & 0.3 & 0.3 \\ 0.2 & 0.2 & 0.3 & 0 & 0.2 \\ 0.1 & 0.1 & 0.3 & 0.2 & 0 \end{pmatrix}, \end{aligned}$$

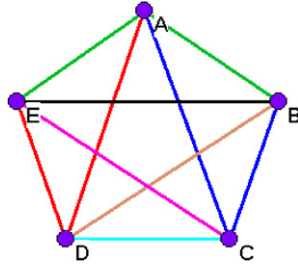


Fig. 1. Complex system of five stocks.

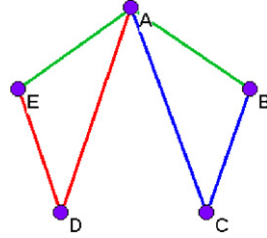


Fig. 2. The forest of all minimal spanning trees.

Step 2. Since $D^{*2} \neq D$, we continue to compute D^{*4} ,

$$D^{*4} = \begin{pmatrix} 0 & 0.1 & 0.3 & 0.2 & 0.1 \\ 0.1 & 0 & 0.3 & 0.2 & 0.1 \\ 0.3 & 0.3 & 0 & 0.3 & 0.3 \\ 0.2 & 0.2 & 0.3 & 0 & 0.2 \\ 0.1 & 0.1 & 0.3 & 0.2 & 0 \end{pmatrix} * \begin{pmatrix} 0 & 0.1 & 0.3 & 0.2 & 0.1 \\ 0.1 & 0 & 0.3 & 0.2 & 0.1 \\ 0.3 & 0.3 & 0 & 0.3 & 0.3 \\ 0.2 & 0.2 & 0.3 & 0 & 0.2 \\ 0.1 & 0.1 & 0.3 & 0.2 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0.1 & 0.3 & 0.2 & 0.1 \\ 0.1 & 0 & 0.3 & 0.2 & 0.1 \\ 0.3 & 0.3 & 0 & 0.3 & 0.3 \\ 0.2 & 0.2 & 0.3 & 0 & 0.2 \\ 0.1 & 0.1 & 0.3 & 0.2 & 0 \end{pmatrix},$$

Step 3. Since $D^{*4} = D^{*2}$, then the SDU of D is D^{*4} and we continue to compute the adjacency matrix Δ that corresponds to the forest of all MSTs of D .

Step 4. To compute Δ , first, we compute $D - D^{*4}$ (matrix subtraction in usual sense).

$$D - D^{*4} = \begin{pmatrix} 0 & 0.1 & 0.3 & 0.2 & 0.1 \\ 0.1 & 0 & 0.3 & 0.4 & 1.7 \\ 0.3 & 0.3 & 0 & 0.6 & 0.5 \\ 0.2 & 0.4 & 0.6 & 0 & 0.2 \\ 0.1 & 1.7 & 0.5 & 0.2 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0.1 & 0.3 & 0.2 & 0.1 \\ 0.1 & 0 & 0.3 & 0.2 & 0.1 \\ 0.3 & 0.3 & 0 & 0.3 & 0.3 \\ 0.2 & 0.2 & 0.3 & 0 & 0.2 \\ 0.1 & 0.1 & 0.3 & 0.2 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.2 & 1.6 \\ 0 & 0 & 0 & 0.3 & 0.2 \\ 0 & 0.2 & 0.3 & 0 & 0 \\ 0 & 1.6 & 0.2 & 0 & 0 \end{pmatrix},$$

Step 5. Then, according to (4),

$$\Delta = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

This matrix Δ indicates that the MST in D is not unique since $N = 6$ exceeds $n - 1 = 4$. It represents the forest in D that can be visualized in Fig. 2.

Actually, there are four MSTs in the forest and each of them has a different network topology compared to the others. Therefore, each of them has different meaning in terms of stocks' interrelationships. That is why the use of a single MST might provide misleading information.

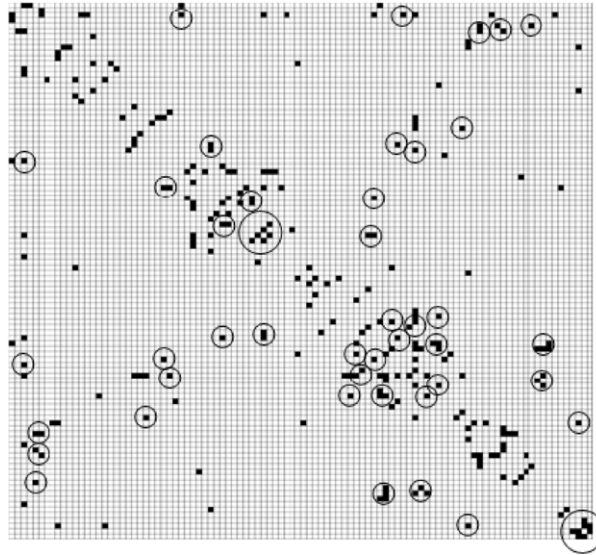


Fig. 3. Dot plot matrix representing the adjacency matrix related to the forest.

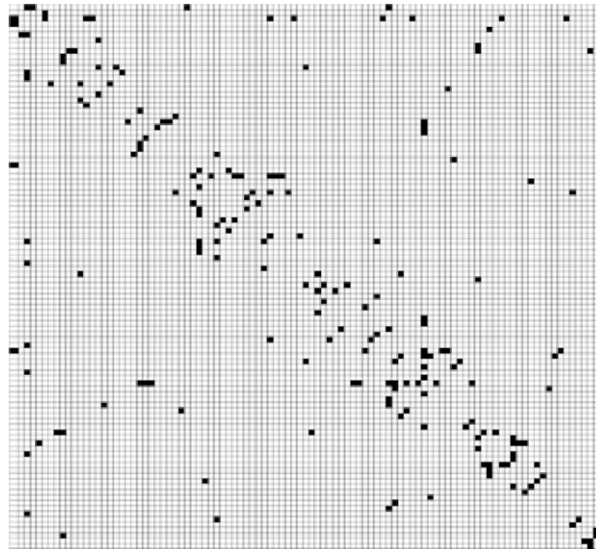


Fig. 4. Dot plot matrix related to an MST issued from Kruskal's algorithm.

5.2. NYSE 100 stocks case

To illustrate the advantages of the proposed filter, compared to the MST-based filter, daily data of the NYSE 100 stock prices from November 18, 2010 until March 28, 2012 will be used here. Those data were downloaded on March 28, 2012 from http://www.nyse.com/about/listed/nyid_components.shtml for the list of the 100 stocks and from <http://finance.yahoo.com/> for daily data of stock price.

First, to simplify the presentation, in Fig. 3 we represent the adjacency matrix Δ related to the forest in the form of a dot plot matrix. The entry (i, j) is dotted if $\delta(i, j) = 1$ and is left blank if $\delta(i, j) = 0$. According to Δ , the number of pairs (i, j) where $i > j$ and $\delta(i, j) = 1$ is $N = 129$. Therefore, since N is greater than $n - 1 = 99$, the MST in the complex system of NYSE 100 stocks is not unique. There are many MSTs in D . In Fig. 3, the non-uniqueness is indicated by circled dots. These circles refer to a set of dots where one or more dots do not appear in the dot plot matrix related to an MST issued from Kruskal's algorithm (see Fig. 4). Furthermore, the convergence speed of the proposed algorithm is quite promising. It needs only $k = 4$ iterations to check the convergence. Thus, $D^{*16} = D^{*8}$, the SDU is D^{*16} .

Second, in Fig. 5 we present the network topology defined by the forest. It is also a topological representation of the dot plot matrix in Fig. 3. Since it is the union of all possible MSTs in D , the MST in Fig. 6 is only a sub-graph of the forest in Fig. 5.

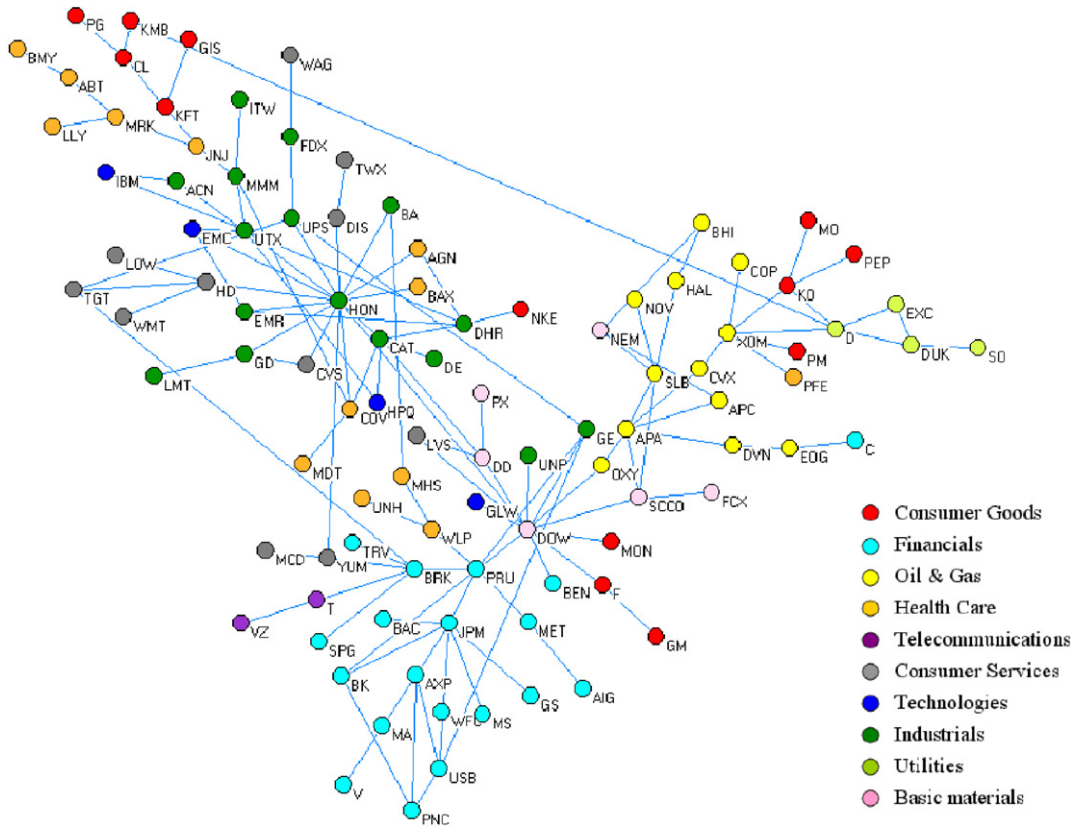


Fig. 5. The forest of all possible MSTs.

From Figs. 5 and 6 we see clearly that in terms of degree centrality measure, for example, the forest-based filter gives a unique degree distribution, while this is not the case if one uses the MST-based filter. Two different MSTs will give two different degree distributions. This centrality measure will be discussed further in the next section.

6. Discussion

Two important issues will be discussed in this section; the topological properties and the construction of the SDU.

6.1. Topological properties

The second example in the previous section shows that the topological properties of stock networks created by an MST such as, for an example, discussed in Tabak et al. [13] are different from those issued by the forest of all MSTs, except when D contains one unique MST. Therefore, with respect to any MST, centrality measures such as degree centrality, betweenness centrality, closeness centrality, and eigenvalue centrality might not be robust. This is not the case if one works based on the forest.

Let us consider, for example, the topological properties in terms of degree centrality (see, e.g., [14–16]), which is representative of market index [2]. The stocks with high degree are more strongly correlated with market index than those with low degree [5,6]. Suppose D contains more than one MST and let M be an MST with adjacency matrix Δ_M and F be the forest of all MSTs. According to M , the degree of stock k is the number of links adjacent to that stock in M or, equivalently, the sum of all elements in the k -th row or column of Δ_M , i.e.,

$$\partial_M(k) = \sum_{j=1}^n \delta_M(k, j). \quad (5)$$

Here $\delta_M(k, j)$ is the element in the k -th row and j -th column of Δ_M . Since M is an MST, we have $\sum_{k=1}^n \partial_M(k) = n - 1$. On the other hand, according to the forest F , the degree of stock k is

$$\partial_F(k) = \max_{M \in F} \{\partial_M(k)\}. \quad (6)$$

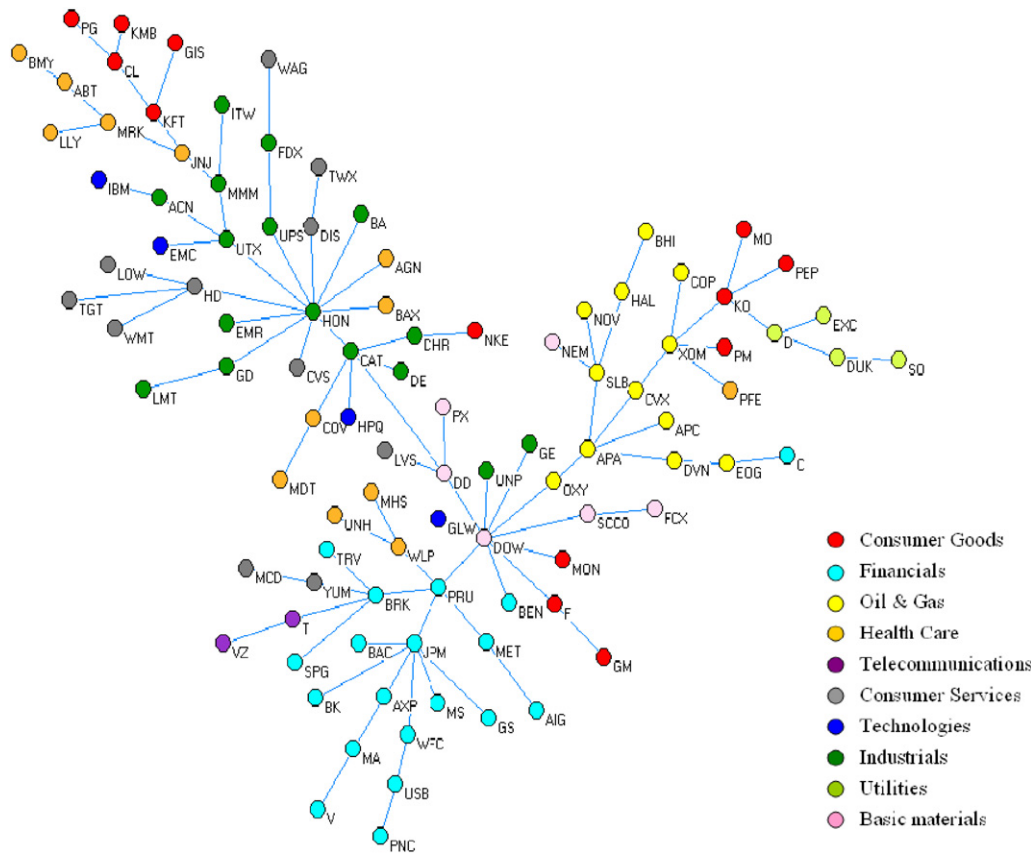


Fig. 6. An MST given by Kruskal's algorithm.

Table 1
Top ten stocks in terms of degree centrality.

Rank	According to the forest	According to MST
1	HON	HON
2	DOW	DOW
3	JPM	JPM
4	PRU	CAT
5	UTX	BRK
6	BRK	PRU
7	CAT	XOM
8	XOM	APA
9	APA	DD
10	DHR	KO

Therefore, if $\partial_M(k)$ in (5) depends on M , $\partial_F(k)$ in (6) does not and, in general, we have $\partial_F(k) \geq \partial_M(k)$ for all $k = 1, 2, \dots, n$. The equality is true if and only if the MST in D is unique.

In Table 1 we list the top ten stocks traded at NYSE, ranked by the proposed filter in the second column and by the MST-based filter in third column. The lists are different from each other. Furthermore, the forest-based filter puts UTX (United Technologies Corp.) and DHR (Danaher Corp.) among the top ten whereas, according to the MST-based filter in Fig. 6, they are respectively at ranks 12 and 34. On the other hand, DD (E.I. DuPont de Nemours & Co.) and KO (Coca Cola Co.) are among the top ten when we use the MST-based filter in Fig. 6 but they are respectively at ranks 13 and 15 according to the forest.

The same situation will also arise if we use other centrality measures. Whatever the centrality measure that we use, the list of stocks ordered from the smallest until the largest ranks given by the proposed filter is different from that given by the MST-based filter. All the above observations illustrate the advantages of the proposed filter.

6.2. SDU construction

To construct an indexed hierarchical tree from a given distance matrix (more generally a dissimilarity matrix) D , it is sufficient to transform D to an ultrametric distance matrix U . For this purpose, there are two types of transformation that

can be found in the literature. The first type gives the ultrametric but not the SDU. Average linkage and complete linkage are examples of this type [17]. The second type gives U in the form of SDU. In this case, to construct U , two techniques can be used;

- (i) The first technique is to construct an MST first. Then, U is the SDU that corresponds to that MST (single linkage). In this case, Prim's algorithm and Kruskal's can be used to find an MST. This is the technique that is usually used in stock networks analysis. MST is to filter the important information contained in stock networks in the form of network topology and SDU is to construct an economic classification in the form of an indexed hierarchical tree.
- (ii) The second technique is to construct U directly from D by using, for example, Johnson's algorithm and Roux's algorithm [17].

It is important to note that, if MST in D is not unique, all MSTs define the same SDU. Furthermore, that SDU is also the SDU associated with the forest. This is a direct consequence of Theorem 2 because, according to the definition of $L(\gamma)$ in that theorem, it might happen that;

- (i) For a given γ in Γ there are two positive integer k and m such that $d(i_k, i_{k+1}) = d(i_m, i_{m+1})$; $k, m = 1, 2, \dots, p$.
- (ii) For two γ_1 in γ_2 in Γ , we have $L(\gamma_1) = L(\gamma_2)$.

7. Concluding remarks

It is very often in practice that a network between stocks contains more than one MST. In this case, the use of an MST to filter important information might be misleading. To handle this problem, we propose to use the forest of all MSTs as a robust filter. As a consequence, with respect to the forest, all centrality measures mentioned in Section 6 are robust.

For practical purposes we present an algorithm to find that forest. It can also be used to detect whether the MST in D is unique or not. MST is unique if and only if the sum of all entries of the adjacency matrix Δ is 2 ($n - 1$). That algorithm involves the following simple matrix operations,

- (i) matrix multiplication in the usual sense but multiplication and summation of two real numbers a and b are defined as $\max\{a, b\}$ and $\min\{a, b\}$, respectively.
- (ii) matrix subtraction in the usual sense.

It needs only k iterations to converge, where $k \leq \frac{\ln(n)}{\ln(2)}$.

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